

HOUSE PRICE PREDICTION

Task 5 — Insights & Summary

This project focused on predicting house prices using machine learning techniques. The Housing Prices Dataset was loaded into Python using Pandas and explored to understand its structure and contents. The dataset contained information about various house characteristics such as area, number of bedrooms, number of bathrooms, stories, parking spaces, air conditioning, furnishing status, and other amenities. The target variable for prediction was the house price.

During the data cleaning stage, the dataset was checked for missing values and duplicate records. No missing values were found in any of the columns, and the dataset was already clean. Categorical variables such as main road, guestroom, basement, air conditioning, and furnishing status were converted into numerical form using one-hot encoding so that machine learning algorithms could process them effectively.

Two machine learning models were trained and evaluated: Linear Regression and Random Forest Regressor. The dataset was divided into training and testing sets using an 80:20 split. Linear Regression achieved a Mean Absolute Error (MAE) of approximately 970,043, a Root Mean Squared Error (RMSE) of approximately 1,324,507, and an R^2 Score of 0.653. Random Forest Regressor achieved an MAE of approximately 1,021,546, an RMSE of approximately 1,400,566, and an R^2 Score of 0.612. Based on these results, Linear Regression performed better on this dataset because it produced lower prediction errors and a higher R^2 score.

Several visualizations were created to better understand the data. The histogram of house prices showed the overall distribution of prices. The correlation heatmap highlighted the relationships between features and house price. Additional charts such as Price vs Area and Average Price by Bedrooms provided further insights into factors affecting house values.

The analysis revealed that features such as area, number of bathrooms, air conditioning, and preferred area have a strong influence on house prices. One interesting observation was that Linear Regression outperformed Random Forest, despite Random Forest often being considered a more advanced model. Based on these findings, real estate businesses should focus on properties with larger areas and better amenities, as these factors contribute significantly to higher market value and customer demand. Some insights are :

1. Features such as area, bathrooms, air conditioning, and preferred area have a strong influence on house prices.
2. Linear Regression performed better than Random Forest Regressor. It achieved an R^2 score of 0.653, meaning it explained about 65.3% of the variation in house prices.
3. The average prediction error was approximately ₹970,043. Although not perfect, the model provides reasonably accurate house price predictions.
4. One surprising observation was that Linear Regression outperformed Random Forest on this dataset, despite Random Forest generally being considered a more powerful model.
5. Based on the analysis, real estate businesses should focus on larger houses with better amenities and preferred locations because these features contribute significantly to higher house prices.